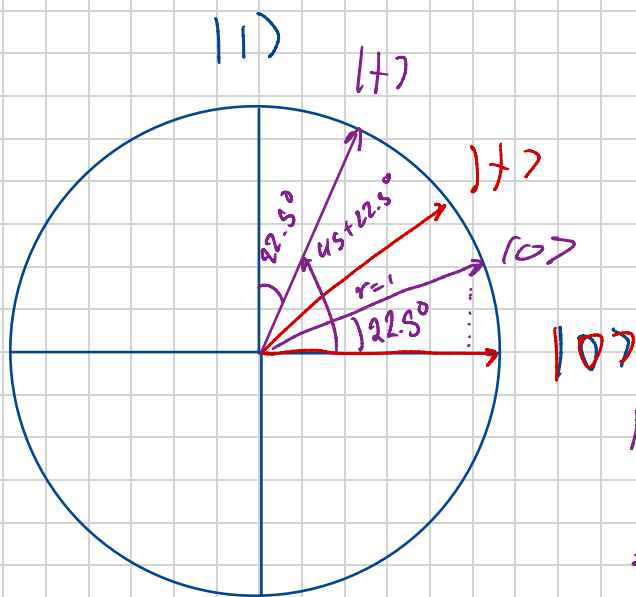


Can we improve on this?

Exercise 4.1 (fairly straightforward). Can you think of a simple way of combining the two strategies above to attain a worst-case success probability of $\frac{3}{4}$?

In fact, there is a better strategy than all of the strategies considered so far.



if we measure now
 $|+\rangle \rightarrow P(0) = \frac{1}{2}, P(1) = \frac{1}{2}$

$|0\rangle \rightarrow |0\rangle$

rot $45/2 = 22.5^\circ$

$|0\rangle$ becomes $\cos(22.5^\circ)|0\rangle + \sin(22.5^\circ)|1\rangle$

where $\cos^2(22.5^\circ) + \sin^2(22.5^\circ) = 1$

$\therefore P(0) = \cos^2(22.5^\circ) = 0.854$

$P(1) = 1 - 0.854$

So worst case = 0.854 success

solved ex 4.2

by accident not 4.1

$|1\rangle$ become $\cos(45^\circ + 22.5^\circ)|0\rangle + \sin(45^\circ + 22.5^\circ)|1\rangle$

$P(1) = \sin^2(45^\circ + 22.5^\circ) = 0.854$

$P(0) = 1 - 0.854$

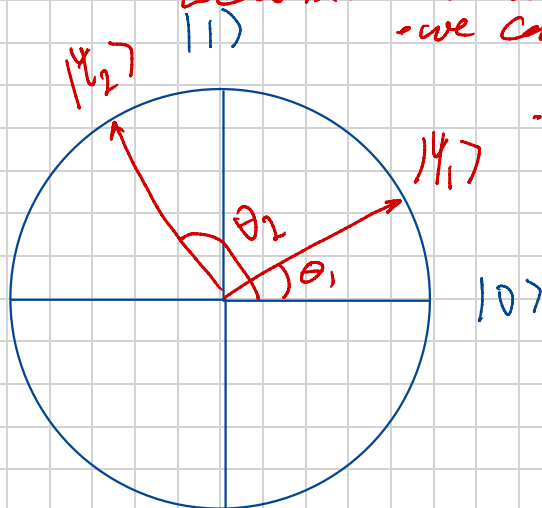
Penny Game: Quantum State Discrimination
 Average case success

$$P_{\text{success}} = P[\text{identified } \psi_1 | \text{given } \psi_1] P[\text{given } \psi_1] + P[\text{ident. } \psi_2 | \text{given } \psi_2] P[\text{given } \psi_2]$$

q_1 P_1 q_2 P_2

• we want to maximize q_1 & q_2 .

• we can only measure once.

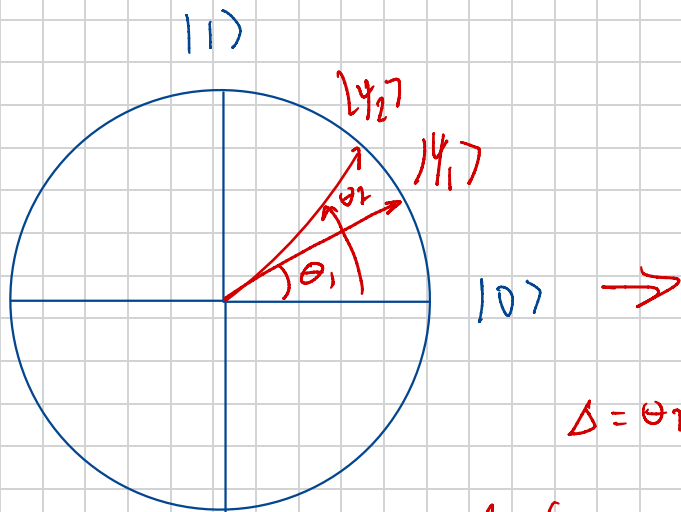
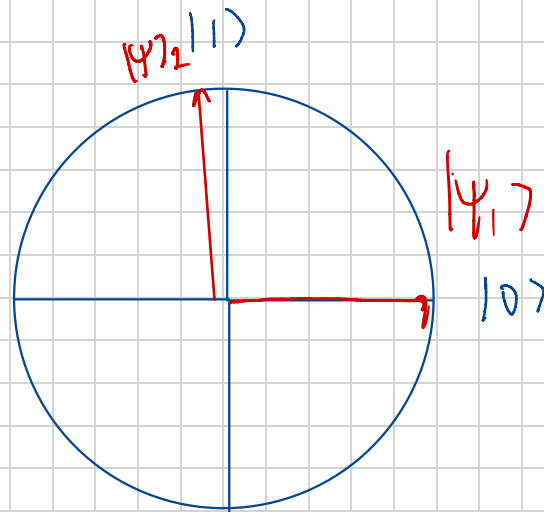
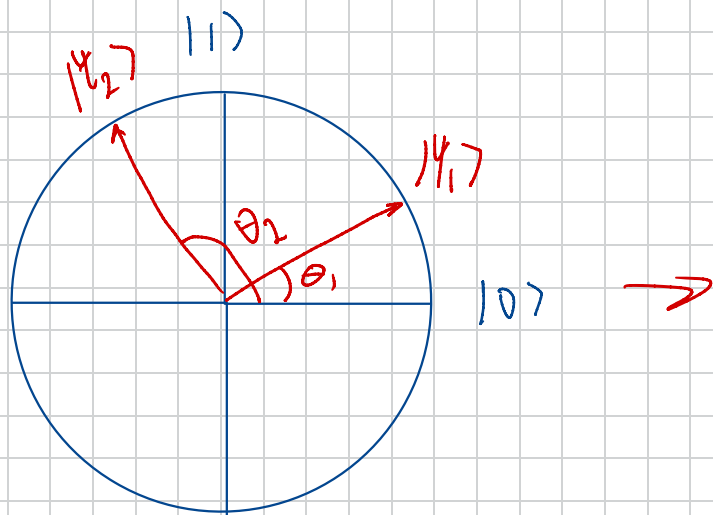


• given $\theta_1, \theta_2, P_1, P_2$

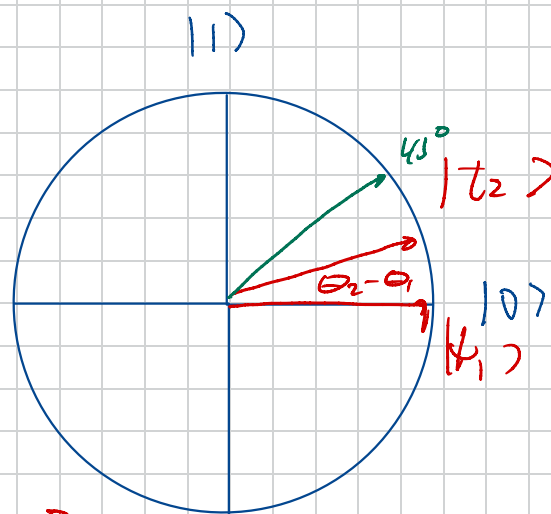
• Start: if $P_1 > P_2$, then

more acceptable to have $q_2 > q_1$

where $\theta_1 < \theta_2$
 not by $-\theta_1$ to

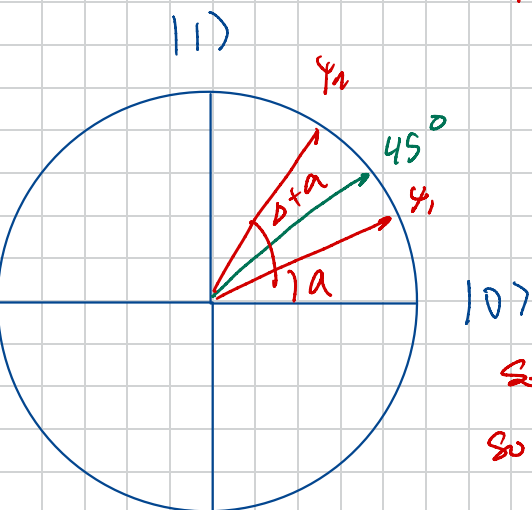


$$\Delta = \theta_2 - \theta_1$$



if $(\theta_2 - \theta_1) \leq 45^\circ$

then rotate $45 - \frac{(\theta_2 - \theta_1)}{2}$ or rotate $45 - \frac{(\theta_2 - \theta_1)}{2} w$ where w is a function of p_1 and p_2 .
 Let's call this extra angle "a"



so $\downarrow a$, prioritizes the $|\psi_1\rangle$ true
 so maybe $\downarrow a$, as $\uparrow p_1$

so now
 $P_{\text{success}}(a) = p_1 \cos^2(a) + p_2 (\sin^2(a + \Delta))$, derive to maximize

→ now
 $P_{\text{success}}(\alpha) = p_1 \cos^2(\alpha) + p_2 \sin^2(\alpha + \Delta)$, derive to maximize

$$P'_{\text{success}}(\alpha) = -p_1 \sin(2\alpha) + p_2 \sin(2\alpha + 2\Delta) = 0$$

Solving gives

$$\alpha_{\text{optimal}} = \frac{1}{2} \arctan 2(p_2 \sin(2\Delta), p_1 - p_2 \cos(2\Delta))$$

This assumes no complex phase; guess
(working on a 2D circle, not a 3D sphere)

- ✓. If/From measurement optimal where
we consider complex relative phases